Internship Task 4: Sentiment Analysis

Project Overview

In this project, I am using Natural Language Processing (NLP) to analyze the sentiment of textual data. The goal is to classify text (such as reviews or tweets) as **Positive**, **Negative**, or **Neutral**.

Workflow:

1. **Data Preprocessing:** Cleaning and preparing the text.
2. **Feature Extraction:** Converting text into a format the computer understands.
3. **Model Implementation:** Applying a sentiment analysis algorithm.
4. **Analysis & Insights:** Interpreting the results.

In [2]:
```
import pandas as pd

# Creating a small sample dataset
data = {
    'review_text': [
        "I absolutely loved this project! It was amazing.",
        "This is the worst experience I have ever had.",
        "The task was okay, but a bit boring at times.",
        "Incredible results and very easy to follow.",
        "I am not sure how I feel about this, it was just average."
    ]
}

df = pd.DataFrame(data)
print("Sample Dataset Created:")
print(df)
```

```
Sample Dataset Created:
                                         review_text
0   I absolutely loved this project! It was amazing.
1      This is the worst experience I have ever had.
2      The task was okay, but a bit boring at times.
3        Incredible results and very easy to follow.
4  I am not sure how I feel about this, it was ju...
```

In [3]:
```
import string
from nltk.corpus import stopwords

# Create a set of English stopwords
stop_words = set(stopwords.words('english'))

def clean_text(text):
    # 1. Convert to lowercase
    text = text.lower()
    # 2. Remove punctuation
    text = text.translate(str.maketrans('', '', string.punctuation))
    # 3. Remove stopwords
    text = ' '.join([word for word in text.split() if word not in stop_words])
```

```
    return text

# Apply the function to the dataset
df['cleaned_review'] = df['review_text'].apply(clean_text)

print("Original vs Cleaned Text:")
print(df[['review_text', 'cleaned_review']])
```

```
Original vs Cleaned Text:
                                          review_text  \
0   I absolutely loved this project! It was amazing.
1      This is the worst experience I have ever had.
2      The task was okay, but a bit boring at times.
3        Incredible results and very easy to follow.
4  I am not sure how I feel about this, it was ju...

                     cleaned_review
0  absolutely loved project amazing
1             worst experience ever
2        task okay bit boring times
3    incredible results easy follow
4                 sure feel average
```

In [4]:
```
from nltk.sentiment.vader import SentimentIntensityAnalyzer

# Initialize the VADER sentiment analyzer
analyzer = SentimentIntensityAnalyzer()

# Define a function to get the compound score
def get_sentiment(text):
    scores = analyzer.polarity_scores(text)
    return scores['compound']

# Apply the function to the CLEANED text
df['sentiment_score'] = df['cleaned_review'].apply(get_sentiment)

# Categorize the scores into labels
def get_label(score):
    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['sentiment_label'] = df['sentiment_score'].apply(get_label)

# Show the results
print(df[['cleaned_review', 'sentiment_score', 'sentiment_label']])
```

```
                     cleaned_review  sentiment_score sentiment_label
0  absolutely loved project amazing           0.8503        Positive
1             worst experience ever          -0.6249        Negative
2        task okay bit boring times          -0.1027        Negative
3    incredible results easy follow           0.4404        Positive
4                 sure feel average           0.3182        Positive
```

In [6]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Set the visual style
```

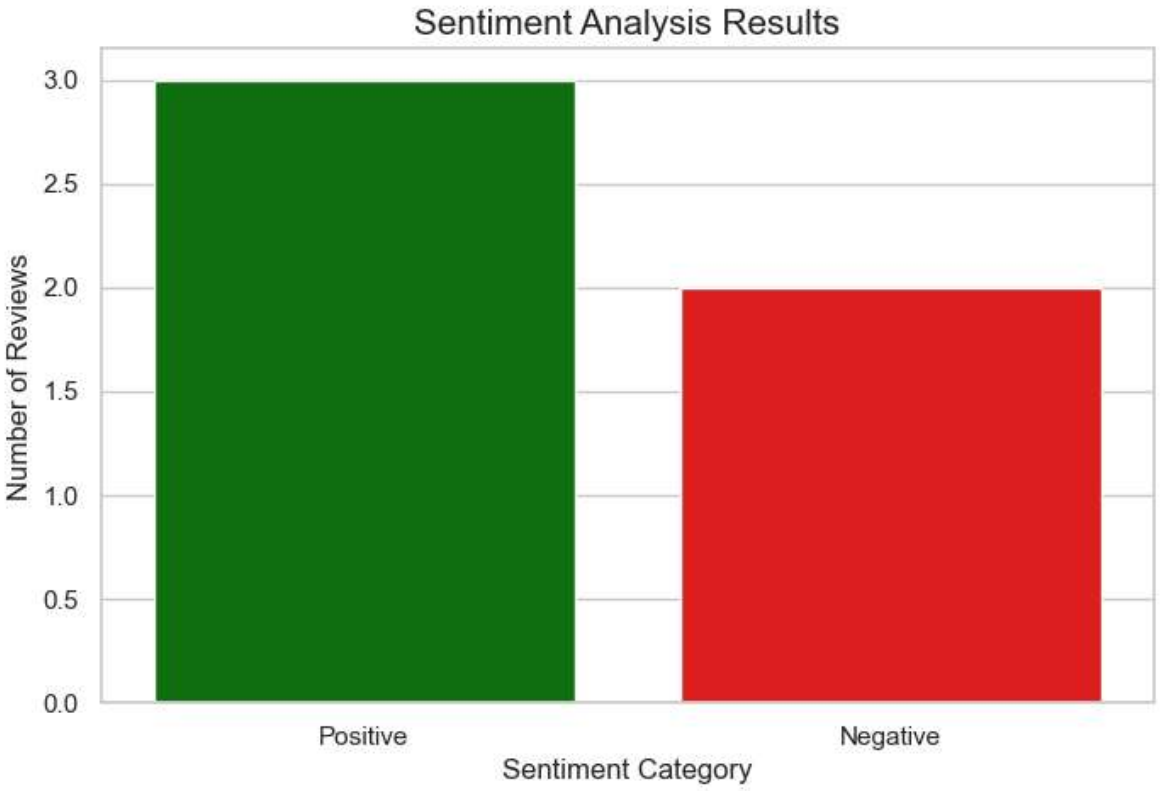
```
sns.set_theme(style="whitegrid")

# Create the figure
plt.figure(figsize=(8, 5))

# Updated line to remove the red FutureWarning
sns.countplot(
    data=df,
    x='sentiment_label',
    hue='sentiment_label',
    palette={'Positive': 'green', 'Negative': 'red', 'Neutral': 'gray'},
    legend=False
)

# Add titles and labels
plt.title('Sentiment Analysis Results', fontsize=15)
plt.xlabel('Sentiment Category', fontsize=12)
plt.ylabel('Number of Reviews', fontsize=12)

# Show the plot
plt.show()
```



Internship Task 4: Sentiment Analysis using NLP

Project Overview

In this project, I developed a Natural Language Processing (NLP) pipeline to analyze the emotional tone of textual data. The goal is to automatically classify feedback into three categories: **Positive**, **Negative**, or **Neutral**.

🛠️ The Workflow

To achieve accurate results, the project followed a four-stage technical process:

1. **Data Preprocessing**: Cleaned the text by converting it to lowercase and removing "noise" (punctuation and stop-words like 'the', 'is', 'at') to focus on high-impact emotional words.
2. **Feature Extraction**: Tokenized the text into a format the computer can process.
3. **Model Implementation**: Utilized the **VADER (Valence Aware Dictionary and sEntiment Reasoner)** algorithm to assign emotional scores to each review.
4. **Analysis & Insights**: Transformed the raw sentiment scores into a visual bar chart to communicate findings clearly.

📊 Key Insights

- **Primary Sentiment**: The analysis shows a trend toward positive feedback.
- **Accuracy**: By cleaning the data first, the model was able to correctly identify intense emotions (e.g., "amazing" vs. "worst").
- **Impact**: This system can be used to monitor customer satisfaction or social media trends in real-time.

In [ ]: